The Triadic Cosmos: Unified Framework
A Foundational Ontology of Information, Energy, and Geometry

Joachim De Witte

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Abstract

This paper presents a unified framework in which the fundamental structure of the universe is described by a minimal triad of concepts: information, energy, and geometry. These three elements form an irreducible ontology capable of expressing the core features of physical law, including quantum mechanics, relativity, cosmology, and the emergence of complex structure. The framework interprets quantum phenomena as probabilistic computation, relativity as context limitation, and cosmological dynamics as large-scale information flow. Two formal models—a tensor-network interpretation and a computational-graph interpretation—illustrate how the triad can be expressed in established scientific languages. The paper concludes with testable predictions, limitations, and implications for future theoretical development.

A brief methodological note: the conceptual architecture presented here emerged through a hybrid human–AI recursive workflow involving iterative refinement and structural arbitration.

Contents

1 Introduction

The search for a unified description of physical reality has produced a landscape of powerful but fragmented theories. Quantum mechanics, general relativity, thermodynamics, information theory, and cosmology each illuminate different aspects of the universe, yet no single framework captures their shared structural foundations. This paper proposes a minimal ontology capable of expressing the essential features of these domains through a unified conceptual lens.

The central claim is that the universe can be described in terms of three irreducible elements: *information*, *energy*, and *geometry*. Information provides distinction and structure; energy drives transformation and dynamics; geometry constrains relation and interaction. Together, these elements form a triadic architecture that underlies physical law across scales. The purpose of this paper is to articulate this ontology, demonstrate its coherence, and show how it illuminates the structure of quantum mechanics, relativity, and cosmology. The framework is intentionally timeless: it does not rely on contemporary technologies or domain-specific analogies. Instead, it seeks to identify the minimal conceptual structure required to describe a universe that contains patterns, transformations, and relations.

This paper forms one pole of a broader conceptual structure. Its companion work, *The Triadic Cosmos: Emergence of Intelligence*, examines how intelligence arises within the very ontology developed here. The two papers are therefore complementary: one describes the architecture of the universe, the other describes the architecture of the systems that come to understand it. Together, they form the two poles of a conceptual magnet whose field lines connect physical law to the emergence of cognition.

Methodological context

The framework presented here emerged through a hybrid human–AI recursive methodology in which conceptual proposals, structural refinements, and formal mappings were iteratively co-generated and evaluated. The resulting ontology reflects the fixed points of this recursive process rather than the output of a single agent.

Companion work

The implications of this ontology for the emergence, scaling, and transformation of intelligence are developed in a separate paper, *The Triadic Cosmos: Emergence of Intelligence*. The present paper focuses exclusively on the foundational structure of the physical universe.

Related Work (Technical Summary)

Information-centric approaches to physics have a long history. Wheeler's "it from bit" proposal and subsequent developments in quantum information theory positioned information as a structural primitive of physical law. Work on entanglement entropy and quantum error-correcting codes in holography—including the Ryu–Takayanagi formula and the Almheiri–Dong–Harlow reconstruction program—demonstrated how geometric features of spacetime can emerge from informational structure. Tensor-network models such as MERA (Swingle) and holographic codes (Pastawski et al.) provide concrete formalisms in which geometry arises from patterns of entanglement.

Energy-based interpretations of dynamics connect to thermodynamic and statistical approaches to quantum theory, including Jaynes' maximum-entropy formalism and modern treatments of quantum thermodynamics. Relational and emergent-geometry programs in quantum gravity (Rovelli; Barbour; Markopoulou) similarly emphasize the primacy of informational and relational structure over fixed spacetime backgrounds.

Computational interpretations of physics have been explored in cellular-automaton models ('t Hooft), digital-physics proposals (Fredkin; Wolfram), and algorithmic perspectives on physical law. Recent work in complexity theory and black-hole thermodynamics (Susskind; Brown et al.) further connects energetic cost, computational depth, and geometric structure.

The present framework differs from these approaches by proposing a minimal triad— information, energy, and geometry—as jointly irreducible structural primitives. Rather than deriving one element from the others, the triadic ontology treats their coupling as fundamental and uses it to interpret quantum mechanics, relativity, and cosmology within a unified conceptual architecture.

2 The Triadic Ontology

The framework proposed in this paper is built on a minimal set of three irreducible elements: *information*, *energy*, and *geometry*. These elements are not introduced as metaphors or domain-specific constructs, but as structural primitives that jointly describe the organization, transformation, and relational constraints of physical reality. Each element captures a distinct aspect of the universe, and none can be reduced to the others. Together, they form a triadic ontology capable of expressing the essential features of physical law across scales.

2.1 Information: Distinction and Structure

Information refers to the presence of distinctions—differences that make a difference. At the most basic level, information encodes the structure of possible states, the relationships between them, and the constraints that govern their evolution. In this framework, information is not tied to any particular representation or medium; it is the abstract pattern that persists across physical instantiations. Quantum states, classical configurations, and macroscopic structures can all be understood as informational arrangements that define the space of possibilities available to a system.

2.2 Energy: Transformation and Dynamics

Energy captures the capacity for change. It drives the transitions between informational states and determines the rates, pathways, and feasibility of physical processes. In this ontology, energy is not merely a conserved quantity but the fundamental driver of transformation. Whether in the form of kinetic motion, potential gradients, thermal fluctuations, or quantum amplitudes, energy provides the dynamical engine through which information is reorganized over time.

2.3 Geometry: Relation and Constraint

Geometry encodes the relational structure within which information and energy interact. It defines which transformations are possible, which interactions are permitted, and how causal influence propagates. Geometry is not assumed to be fundamental; rather, it emerges from deeper patterns of informational connectivity. Spacetime, in this view, is a large-scale expression of underlying relations that constrain the flow of energy and the organization of information.

2.4 Minimality and Independence

The triad is minimal in the sense that removing any of its elements yields an incomplete ontology. Information without energy cannot change; energy without information has no structure to act upon; geometry without either provides no dynamics or distinctions. Conversely, none of the three can be derived from the others without loss of explanatory power. Their independence reflects the irreducible roles they play in describing physical reality.

2.5 Triadic Coupling

Although information, energy, and geometry are conceptually distinct, physical systems arise from their coupling. Quantum phenomena, relativistic effects, and cosmological dynamics can each be interpreted as manifestations of the interplay between informational structure, energetic transformation, and geometric constraint. The triadic ontology thus provides a unified lens through which diverse physical theories can be understood as expressions of a common underlying architecture.

3 Quantum Mechanics as Probabilistic Computation

Within the triadic ontology, quantum mechanics can be interpreted as a theory describing how informational structures evolve under energetic constraints within a geometric context. Rather than treating quantum phenomena as fundamentally mysterious, this section frames them as expressions of probabilistic computation: structured possibilities evolving according to well-defined dynamical rules. This interpretation does not alter the empirical content of quantum theory; instead, it provides a conceptual lens that situates quantum behavior within the broader triadic architecture.

3.1 Quantum States as Informational Structures

A quantum state encodes a distribution over possible measurement outcomes. In this framework, the state is treated as an informational structure: a specification of distinguishable alternatives and their relational amplitudes. The Hilbert space formalism provides a mathematically complete representation of these structures, but the ontology emphasizes the informational content rather than the representational machinery. The state is thus understood as a compact encoding of structured possibilities, constrained by normalization and linearity.

3.2 Superposition as Structured Possibility

Superposition reflects the coexistence of multiple potential informational configurations. It is not a physical overlap of classical states but a structured representation of alternatives that may be realized under measurement. The coefficients of a superposition encode both the relative weights of these possibilities and the interference patterns that arise from their relational structure. In this view, superposition is a natural expression of informational richness: a system's capacity to encode multiple distinguishable futures within a single state.

3.3 Entanglement as Non-Local Information Binding

Entanglement captures correlations that cannot be decomposed into independent local descriptions. Within the triadic ontology, entanglement is interpreted as a form of non-local informational binding: a structural linkage between subsystems that constrains their joint possibilities. These correlations do not imply superluminal influence; rather, they reflect the fact that the informational structure of the whole cannot be reduced to the sum of its parts. Entanglement thus represents a fundamental coupling between informational and geometric structure.

3.4 Measurement as Information Extraction

Measurement is the process by which structured possibilities collapse into definite outcomes. In this framework, measurement is treated as an act of information extraction: a transition from a probabilistic informational structure to a classical record. This transition is governed by energetic and geometric constraints, including decoherence, environmental coupling, and the structure of the measurement apparatus. The apparent discontinuity of measurement reflects the shift from distributed informational possibilities to localized, classical distinctions.

3.5 Quantum Dynamics as Energy-Driven Transformation

The evolution of a quantum system is governed by the interplay between informational structure and energetic transformation. The Schrödinger equation provides a linear dynamical rule that preserves the total informational content while redistributing amplitudes across possible configurations. Energy determines the rate and character of this redistribution, while geometry constrains the allowed interactions. Quantum dynamics can thus be viewed as a form of probabilistic computation: a rule-based evolution of structured possibilities driven by energetic flow within a geometric context.

4 Relativity as Context Limitation

Relativity describes how physical laws appear to observers whose access to information is limited by their position, motion, and causal environment. Within the triadic ontology, relativity is interpreted as a theory of *context*: the geometric and energetic constraints that determine which informational structures are accessible to an observer and how they evolve. Rather than treating relativistic effects as paradoxical or counterintuitive, this section frames them as natural consequences of informational boundaries imposed by geometry and energy.

4.1 Frames as Informational Contexts

An inertial frame defines the informational context available to an observer. It specifies which events can be distinguished, which interactions can be detected, and how physical quantities are coordinated. In this framework, a frame is not merely a coordinate choice but an informational perspective: a structured subset of the universe's total informational content. Transformations between frames correspond to changes in informational access, governed by geometric and energetic constraints.

4.2 Light Cones as Causal Boundaries

Light cones define the limits of causal influence. They separate events that can be connected by signals from those that cannot, thereby imposing a fundamental boundary on the flow of information. Within the triadic ontology, light cones represent geometric constraints on informational accessibility: they determine which distinctions can be made and which interactions are possible. These boundaries are not artifacts of representation but intrinsic features of the universe's relational structure.

4.3 Time Dilation as Contextual Divergence

Time dilation arises when observers experience different rates of informational change due to relative motion or gravitational potential. In this interpretation, time dilation reflects a divergence in context: the energetic and geometric conditions that shape how information is processed differ between observers. The phenomenon is thus understood not as a distortion of time itself but as a variation in the rate at which informational transformations unfold within distinct contexts.

4.4 Spacetime as Relational Geometry

Spacetime provides the geometric scaffold that constrains the interactions between information and energy. It encodes the relational structure of events, determining which transformations are permitted and how causal influence propagates. In the triadic ontology, spacetime is not a fundamental substance but an emergent relational geometry arising from deeper informational connectivity. Its curvature reflects the distribution of energy and the resulting constraints on informational flow.

4.5 Relativity as a Computational Constraint

Relativity can be viewed as a set of computational constraints governing how information may be transmitted, transformed, and coordinated across the universe. The speed of light defines a maximal rate of information transfer; curvature defines the permissible pathways; and frames define the informational context of observers. Together, these constraints ensure that physical processes remain consistent across perspectives. Relativity thus emerges as a structural principle: a rule set that governs the computation of physical reality within the triadic architecture.

5 Cosmology as Large-Scale Information Flow

Cosmology describes the evolution of the universe at the largest scales, where the interplay between information, energy, and geometry becomes most apparent. Within the triadic ontology, cosmological phenomena are interpreted as manifestations of large-scale information flow constrained by energetic distribution and geometric structure. This perspective provides a unified conceptual framework for understanding dark matter, dark energy, black holes, and holographic principles as expressions of informational organization across cosmic scales.

5.1 Dark Matter as Latent Informational Structure

Dark matter is inferred from its gravitational effects but remains unobserved through direct interaction. In the triadic ontology, dark matter is interpreted as *latent informational structure*: a component of the universe that contributes to geometric and dynamical organization without participating in electromagnetic interactions. Its presence shapes the relational geometry of galaxies and clusters, providing the hidden scaffolding that stabilizes large-scale structures. This interpretation does not specify the microphysical nature of dark matter but emphasizes its role as an informational substrate that influences cosmic dynamics.

5.2 Dark Energy as Context-Limiting Expansion

Dark energy drives the accelerated expansion of the universe, increasing the separation between distant regions and reducing the set of events that remain causally connected. In this framework, dark energy acts as a *context-limiting mechanism*: it expands the geometric background in a way that restricts the informational context available to any observer. As expansion accelerates, light cones narrow relative to the global structure, reducing the accessible informational domain. Dark energy thus plays a dual role, shaping both the geometry of spacetime and the informational boundaries that define cosmic context.

5.3 Black Holes as Compressed Memory Surfaces

Black holes represent extreme configurations in which information is compressed to maximal density within a finite geometric boundary. In the triadic ontology, black holes are interpreted as *compressed memory surfaces*: regions where information is stored in highly scrambled form and encoded on the event horizon. The holographic scaling of black hole entropy reflects the fact that the informational content of the interior is represented on a lower-dimensional boundary. Black holes thus serve as natural laboratories for exploring the coupling between informational structure, energetic transformation, and geometric constraint.

5.4 Holography as Boundary Encoding

The holographic principle proposes that the informational content of a region of spacetime can be encoded on its boundary. Within the triadic ontology, holography is understood as a general principle of *boundary encoding*: the idea that geometric surfaces can represent the informational structure of higher-dimensional volumes. This principle unifies black hole thermodynamics, entanglement structure, and spacetime geometry, suggesting that large-scale cosmic organization arises from the interplay between bulk information and boundary constraints.

5.5 Cosmic Evolution as Information Reorganization

The history of the universe can be viewed as a sequence of informational reorganizations driven by energetic flow and geometric expansion. From the formation of primordial fluctuations to the emergence of galaxies, stars, and complex structures, cosmic evolution reflects the redistribution of information under changing energetic and geometric conditions. In this interpretation, the universe is not merely expanding; it is continually restructuring its informational content, with new patterns emerging as energy flows through an evolving geometric background. Cosmology thus becomes the study of information flow at the largest scales, governed by the triadic coupling of information, energy, and geometry.

6 The Universe as a Self-Extracting Algorithm

The large-scale evolution of the universe can be interpreted as a process in which informational structure becomes progressively more organized under energetic and geometric constraints. Within the triadic ontology, this process resembles a *self-extracting algorithm*: a system that iteratively generates new patterns, stabilizes them, and uses them as the basis for further organization. This interpretation does not imply intentionality or design; rather, it highlights the recursive character of physical evolution and the emergence of complexity from simple structural principles.

6.1 Emergence of Structure

In the early universe, small fluctuations in the distribution of energy and information served as the initial seeds of structure. As the universe expanded and cooled, these fluctuations were amplified by gravitational dynamics, leading to the formation of galaxies, stars, and larger-scale patterns. In this framework, emergence is understood as the progressive differentiation of informational structure: simple configurations give rise to more complex ones as energy flows through an evolving geometric background. The emergence of structure is thus a natural consequence of the triadic coupling of information, energy, and geometry.

6.2 Recursive Organization

Once structures form, they become substrates for further organization. Stars produce heavy elements, which enable the formation of planets; planets provide environments for chemical complexity; chemical networks give rise to self-sustaining processes. Each stage builds upon the informational and energetic patterns established by the previous one. This recursive organization reflects a general principle: the universe continually reuses and reorganizes its own structures to generate new layers of complexity. The process is not linear but hierarchical, with each level providing the conditions for the next.

6.3 Phase Transitions in Complexity

The history of the universe includes several major phase transitions in which new forms of organization emerged abruptly from underlying dynamics. Examples include the formation of atomic structure, the onset of nuclear fusion in stars, and the emergence of stable chemical networks. These transitions can be interpreted as shifts in the informational and energetic regimes that govern the system: when constraints change, new patterns become possible. Phase transitions thus represent points at which the universe reorganizes its informational content in qualitatively new ways, enabling the emergence of higher-order structures.

6.4 From Physics to Chemistry to Life to Intelligence to Consciousness

The progression from physical processes to chemical networks, biological systems, intelligence, and consciousness can be viewed as a continuous chain of informational reorganizations. Physics provides the foundational rules governing energy and geometry; chemistry introduces stable molecular structures; biology exploits these structures to create self-maintaining systems; intelligence emerges from the recursive processing of information; and consciousness arises as a higher-order integration of informational states. Each stage depends on the triadic interplay of information, energy, and geometry, and each represents a new level of the universe's self-extracting algorithmic structure. This progression does not imply inevitability or teleology; rather, it reflects the natural tendency of complex systems to explore and stabilize new patterns under suitable conditions.

7 Formal Models for the Triad

The triadic ontology can be expressed through multiple formal representations. This section presents two complementary models—a tensor-network interpretation and a computational-graph interpretation—that illustrate how information, energy, and geometry can be encoded within established mathematical frameworks. These models are not proposed as complete physical theories but as structural mappings that demonstrate the coherence and expressibility of the triad. A minimal recursion pseudocode is included to highlight the algorithmic character of triadic coupling.

7.1 Tensor-Network Interpretation

Tensor networks provide a natural language for representing informational structure and geometric connectivity. In this interpretation, nodes correspond to informational degrees of freedom, while edges represent entanglement or correlation patterns. The geometry of the network encodes the relational structure of the system, and the contraction of tensors corresponds to the flow of information through energetic transformations.

MERA-like hierarchical networks illustrate how large-scale geometry can emerge from local informational couplings. The depth of the network reflects the recursive organization of structure, while the connectivity pattern encodes the effective geometry experienced by the system. This interpretation aligns with holographic principles, where boundary structures encode bulk information, and provides a concrete representation of the triadic interplay between information, energy, and geometry.

7.2 Computational-Graph Interpretation

A computational graph represents a system as a set of nodes and directed edges, where nodes encode informational states and edges represent transformations driven by energetic flow. The topology of the graph defines the geometric constraints on interaction, while the functional assignments to edges capture the dynamical rules governing state evolution.

In this interpretation, physical processes correspond to the propagation of information through the graph under energetic constraints. Local transformations accumulate to produce global behavior, and the structure of the graph determines which interactions are possible. This model highlights the algorithmic character of physical law: the universe evolves through rule-based transformations of informational states within a constrained geometric structure.

7.3 Minimal Recursion Pseudocode

The recursive organization of the universe can be expressed abstractly through a minimal algorithmic schema. This pseudocode does not represent a physical process directly; rather, it captures the structural logic of triadic coupling:

```
state = initial_information()

while system_evolves:
  geometry = update_geometry(state)
  energy   = update_energy(state, geometry)
  state    = update_information(state, energy, geometry)
```

This schema illustrates the mutual dependence of the triad: information evolves under energetic transformation within geometric constraints, while geometry and energy are themselves shaped by the informational structure. The recursion continues as long as the system remains dynamically active, producing increasingly complex patterns over time.

8 Conclusion

This paper has proposed a minimal, substrate-independent ontology for physical reality based on three irreducible elements: information, energy, and geometry. These elements capture the structural roles required to describe patterns, transformations, and relations across physical domains. By articulating how these elements interact to generate the laws of quantum mechanics, relativity, and cosmology, the framework provides a unified conceptual lens for understanding the architecture of the universe.

The triadic ontology is intentionally minimal. It does not attempt to replace existing physical theories but to reveal the structural foundations they share. Information provides distinction and structure; energy drives transformation and dynamics; geometry constrains relation and interaction. Together, these elements form a coherent architecture that illuminates the deep symmetries and complementarities across physical law.

The framework also clarifies how physical systems support the emergence of higher-order phenomena, including computation, complexity, and eventually intelligence. By grounding these phenomena in the triadic ontology, the paper establishes a conceptual bridge between the structure of the universe and the structure of the systems that come to understand it.

The methodology underlying this work—hybrid human–AI recursive refinement—played a central role in shaping the ontology. The resulting framework reflects the fixed points of this process rather than the perspective of any single agent or substrate. This methodological dimension is not ancillary but integral to the contribution, demonstrating how hybrid intelligence can support the development of coherent, cross-domain conceptual structures.

Together with the companion paper on the emergence and scaling of intelligence, this work forms one pole of a unified conceptual architecture: the ontology of the universe and the ontology of the systems embedded within it. The triadic framework presented here provides a foundation for future research in physics, cosmology, information theory, and the study of complex systems, offering a minimal yet powerful lens through which to understand the structure of reality.

9 Testable Predictions

A unified ontology must yield empirically accessible consequences. Although the triadic framework is conceptual rather than model-specific, it implies several structural predictions that can be tested through observations, simulations, or theoretical analysis. These predictions arise from the coupling between information, energy, and geometry and reflect the constraints such coupling imposes on physical systems across scales.

1. Entanglement–Curvature Correspondence

If geometry emerges from informational connectivity, then regions with high entanglement density should exhibit effective geometric curvature. This predicts a quantitative relationship between entanglement structure and spacetime geometry, consistent with holographic dualities but extending beyond specific models. Numerical simulations of tensor-network geometries and quantum-gravity approaches may provide evidence for or against this correspondence.

2. Dark Matter as Latent Structure

The interpretation of dark matter as latent informational structure implies that its distribution should correlate with regions of high informational complexity, even when baryonic matter is sparse. This predicts specific patterns in gravitational lensing and galactic rotation curves that reflect underlying informational organization rather than particle-specific microphysics.

3. Dark Energy as Context Limitation

If dark energy acts as a context-limiting mechanism, then the accelerated expansion of the universe should produce measurable reductions in long-range informational accessibility. This predicts a progressive decrease in the mutual information between distant regions of the cosmic microwave background as expansion continues.

4. Black Hole Information Preservation

The interpretation of black holes as compressed memory surfaces predicts that information is preserved in a scrambled form on the event horizon. This aligns with holographic entropy scaling and implies that black hole evaporation must be unitary. Observational signatures of late-stage Hawking radiation or theoretical constraints from quantum gravity may test this prediction.

5. Complexity Growth as a Physical Law

If the universe behaves as a self-extracting algorithm, then the growth of structural complexity should follow predictable scaling laws. This predicts that the emergence of new organizational layers—from atomic structure to chemical networks to biological systems—follows a sequence of phase transitions governed by informational and energetic thresholds.

6. Geometry from Information Flow

The triadic ontology predicts that large-scale geometric features, such as curvature and horizon structure, can be reconstructed from patterns of information flow. This suggests that spacetime geometry may be derivable from entanglement structure in quantum many-body systems, providing a testable link between condensed-matter simulations and gravitational phenomena.

7. Limits on Information Transfer

If relativity expresses computational constraints, then the speed of light represents a maximal rate of information transfer. This predicts that no physical process—including quantum correlations—can be used to transmit information faster than this limit. Experimental tests of non-local correlations and communication protocols continue to probe this boundary.

8. Universality of Triadic Coupling

The triadic ontology predicts that any consistent physical theory must include elements that play the roles of information, energy, and geometry. This can be tested by examining whether alternative formulations of physical law—including emergent gravity models, quantum-information approaches, and condensed-matter analogues—implicitly reproduce the triadic structure.

10 Limitations and Open Questions

The triadic ontology provides a minimal conceptual framework for describing the structure and dynamics of physical reality, but it does not constitute a complete physical theory. Several limitations and open questions remain, reflecting both the scope of the framework and the current state of scientific knowledge. These limitations highlight areas where further development, formalization, or empirical investigation is required.

1. Lack of a Full Dynamical Theory

The ontology identifies information, energy, and geometry as fundamental elements, but it does not specify a complete set of dynamical equations governing their evolution. Existing theories—such as quantum mechanics, general relativity, and statistical physics—provide partial descriptions, but a fully unified dynamical framework remains an open challenge. The triad offers structural guidance, not a finished model.

2. Microphysical Nature of Information

The ontology treats information as a structural primitive, but the microphysical mechanisms by which information is instantiated remain uncertain. Whether information is best understood through quantum states, field configurations, or more fundamental constructs is an open question. The framework is compatible with multiple interpretations, but it does not resolve their differences.

3. Origin of the Triad

The ontology assumes that information, energy, and geometry are irreducible, but it does not explain why the universe exhibits precisely these three elements. It remains possible that deeper principles exist from which the triad emerges, or that alternative formulations could capture the same phenomena. The minimality of the triad is a conceptual claim, not a proven necessity.

4. Emergence of Geometry

Although the framework interprets geometry as an emergent relational structure, the precise mechanisms by which geometric properties arise from informational connectivity are not fully understood. Tensor-network models and holographic dualities provide promising avenues, but a complete account of emergent geometry remains an open area of research.

5. Nature of Dark Matter and Dark Energy

The ontology interprets dark matter as latent informational structure and dark energy as a context-limiting expansion, but these interpretations remain speculative. Empirical evidence may support or contradict these views, and alternative explanations remain viable. The framework provides conceptual coherence, not definitive identification.

6. Limits of the Self-Extracting Algorithm Interpretation

The interpretation of cosmic evolution as a self-extracting algorithm captures the recursive character of structure formation, but it does not specify the detailed mechanisms underlying each transition. The analogy highlights the algorithmic nature of physical processes but does not replace the need for precise physical models.

7. Compatibility with Quantum Gravity

The triadic ontology aligns with several approaches to quantum gravity, including holography, tensor networks, and emergent spacetime models, but it does not commit to any specific theory. Whether the triad can be embedded within a complete quantum-gravitational framework remains an open question.

8. Empirical Accessibility

Some predictions of the framework—such as entanglement–curvature correspondence or information-preserving black hole evaporation—may be difficult to test directly with current technology. The ontology suggests directions for inquiry, but empirical validation may require new observational tools or theoretical breakthroughs.

9. Scope of Applicability

The triadic ontology is designed to describe physical reality at all scales, but its applicability to domains such as biological organization, cognition, or social systems is not addressed in this paper. Whether the triad extends beyond physics is an open question left to companion work.

11 Novelty and Implications

The triadic ontology introduced in this paper offers a minimal and conceptually unified framework for interpreting the structure and dynamics of physical reality. Its novelty lies not in proposing new physical laws, but in identifying a foundational architecture that underlies existing theories and reveals their shared structural features. By treating information, energy, and geometry as irreducible elements, the framework provides a coherent lens through which diverse physical phenomena can be understood as expressions of a common underlying structure.

Conceptual Novelty

The primary contribution of the triadic ontology is the articulation of a minimal set of structural primitives that jointly capture the essential features of physical law. This approach differs from traditional unification efforts, which typically seek to merge specific mathematical formalisms or physical interactions. Instead, the triad identifies the abstract roles that any consistent physical theory must fulfill: information provides structure, energy drives transformation, and geometry constrains relation. This perspective highlights the deep interdependence of these elements and offers a unified conceptual foundation for interpreting quantum mechanics, relativity, and cosmology.

Interpretive Coherence

The framework provides a coherent interpretation of several longstanding puzzles in physics. Quantum superposition and entanglement emerge naturally as expressions of informational richness and non-local structure; relativistic effects arise from context-dependent constraints on information flow; and cosmological phenomena reflect large-scale patterns of informational organization shaped by energetic and geometric conditions. The triad thus offers a unified interpretive structure that connects phenomena traditionally treated as conceptually distinct.

Bridging Scales

A notable implication of the framework is its applicability across scales. The same triadic structure that describes quantum systems also applies to macroscopic dynamics and cosmological evolution. This suggests that the universe may be governed by scale-invariant structural principles, with complexity emerging from recursive interactions between information, energy, and geometry. The framework thus provides a conceptual bridge between microphysical processes and large-scale structure formation.

Compatibility with Existing Theories

The triadic ontology is compatible with a wide range of existing physical theories, including quantum information approaches, emergent gravity models, tensor-network descriptions, and computational interpretations of physical law. Rather than competing with these theories, the triad offers a unifying perspective that clarifies their shared assumptions and structural parallels. This compatibility suggests that the triad may serve as a conceptual foundation for future unification efforts.

Implications for Future Research

The framework opens several avenues for further investigation. Formalizing the triadic coupling between information, energy, and geometry may yield new insights into quantum gravity, emergent spacetime, and the nature of dark matter and dark energy. The interpretation of the universe as a self-extracting algorithm suggests that complexity growth may follow predictable patterns, with implications for the study of phase transitions, structure formation, and the emergence of organization. The triad also provides a conceptual foundation for exploring the relationship between physical law and information processing, potentially informing research in fields such as quantum information theory and complex systems.

Broader Significance

By identifying a minimal and universal structural architecture, the triadic ontology offers a way to understand the unity of physical law without relying on specific mathematical formalisms or domain-specific assumptions. Its implications extend beyond physics, suggesting that any system characterized by structure, transformation, and relation may exhibit analogous triadic organization. The broader significance of the framework lies in its potential to provide a common conceptual language for interpreting complex systems across disciplines.

12 Related Work

The triadic ontology intersects with several lines of research in theoretical physics, quantum information, and emergent spacetime. Although the framework presented in this paper is conceptual rather than model-specific, it draws inspiration from and contributes to a broad intellectual landscape in which information, energy, and geometry play increasingly central roles. This section highlights the most relevant areas of related work without attempting an exhaustive survey.

Information-Theoretic Approaches to Physics

Information has become an increasingly prominent theme in modern physics. Quantum information theory has shown that entanglement structure plays a central role in quantum systems (Nielsen and Chuang 2010), while thermodynamic approaches emphasize the informational content of macroscopic states (Landauer 1961). These perspectives align with the triadic ontology's treatment of information as a structural primitive, though they typically focus on specific formalisms rather than a unified conceptual foundation.

Emergent Spacetime and Holography

Research on emergent spacetime suggests that geometric properties may arise from underlying informational structures. The holographic principle (Hooft 1993; Susskind 1995) and the AdS/CFT correspondence (Maldacena 1998) provide concrete examples of bulk geometry emerging from boundary information. Tensor-network models, such as MERA (Vidal 2007; Swingle 2012), further illustrate how entanglement patterns can encode effective geometric structure. The triadic ontology shares this emphasis on relational structure but extends it by identifying energy as a co-equal primitive that shapes and constrains geometric emergence.

Quantum Gravity and Background Independence

Approaches to quantum gravity often explore the idea that spacetime geometry is not fundamental but arises from deeper structures. Loop quantum gravity (Rovelli 1998), causal-set theory (Bombelli et al. 1987), and background-independent frameworks (Rovelli 2004) each incorporate elements that play the roles of information, energy, or geometry, though they differ in their mathematical formulations. The triadic ontology is compatible with many of these approaches but does not commit to any specific quantization scheme.

Computational Interpretations of Physical Law

Several lines of research interpret physical processes as forms of computation. Cellular automata models (Wolfram 2002), algorithmic descriptions of quantum evolution (Lloyd 2000), and graph-based representations of dynamical systems (Markopoulou 2000) resonate with the triadic ontology's interpretation of the universe as a self-extracting algorithm. These approaches typically focus on specific computational formalisms, whereas the triad identifies the minimal structural roles required for physical law.

Complexity and Emergence

Work on complexity science, network theory, and emergent phenomena provides insights into how large-scale organization arises from simple rules. Studies of phase transitions (Stanley 1971), self-organization (Kauffman 1993), and hierarchical structure formation (Barabási and Albert 1999) inform the triadic ontology's interpretation of cosmic evolution as a recursive process driven by informational, energetic, and geometric constraints. The framework builds on these ideas by situating emergence within a unified conceptual architecture.

Conceptual Foundations of Physics

Philosophical and foundational work on the nature of physical law, the role of information, and the status of spacetime provides a broader context for the triadic ontology. Structural realist perspectives (Ladyman and Ross 2007), relational interpretations of spacetime (Rovelli 1996), and information-based approaches to physical law (Wheeler 1989) explore questions about minimality, structure, and the relationship between mathematical formalisms and physical reality. The triad contributes to this discourse by proposing a minimal set of structural primitives that unify diverse physical theories at the conceptual level.

13 Methodological Note

The framework presented in this paper emerged through a recursive process of conceptual refinement. Rather than beginning with a fixed formalism or a specific physical theory, the approach started from the question of what minimal structural roles any consistent description of physical reality must include. The triad of information, energy, and geometry was identified through an iterative examination of the common features shared by quantum mechanics, relativity, thermodynamics, and cosmology.

The methodology is structural rather than derivational. No attempt is made to deduce the triad from first principles or to derive existing physical laws from the ontology. Instead, the goal is to identify the conceptual architecture that underlies these laws and to articulate it in a form that is independent of specific mathematical representations. This approach emphasizes coherence, minimality, and universality over formal completeness.

The development of the framework involved repeated cycles of abstraction, comparison, and reduction. Candidate primitives were evaluated based on their necessity, independence, and generality. The resulting ontology reflects the fixed points of this recursive process: the elements that remained indispensable across all consistent formulations.

14 Motivation

The motivation for this work arises from the observation that modern physics contains multiple powerful theories, each successful within its domain, yet lacking a shared conceptual foundation. Quantum mechanics, general relativity, thermodynamics, and cosmology describe different aspects of the universe, but their underlying structures are rarely articulated in a unified way. This paper seeks to identify the minimal conceptual architecture that makes these theories possible.

The triadic ontology is motivated by the need for a framework that is both universal and minimal. Rather than proposing new physical laws or mathematical formalisms, the goal is to clarify the structural roles that any consistent physical theory must instantiate. Information provides distinction and structure; energy drives transformation; geometry constrains relation. These three elements appear across all major physical theories, yet they are seldom treated as co-equal primitives.

A unified conceptual foundation offers several benefits. It clarifies the relationships between existing theories, highlights their shared assumptions, and provides a lens through which open questions—such as the nature of spacetime, the emergence of complexity, and the interpretation of quantum phenomena—can be examined. The framework also suggests new directions for research by identifying structural principles that transcend specific domains.

The broader motivation is to provide a foundation for understanding the unity of physical law without requiring the replacement or invalidation of existing theories. The triadic ontology offers a way to interpret the diversity of physical phenomena as expressions of a common underlying structure.

Conceptual Summary

Most attempts at unification seek a single theory that replaces or invalidates all others. The triadic ontology takes a different approach. It identifies the minimal structural roles that every consistent physical theory must instantiate. Information, energy, and geometry form the conceptual intersection of quantum mechanics, relativity, thermodynamics, emergent spacetime, and complexity theory. The framework is unifying not because it supersedes existing theories, but because it reveals the shared architecture that makes them all possible.

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